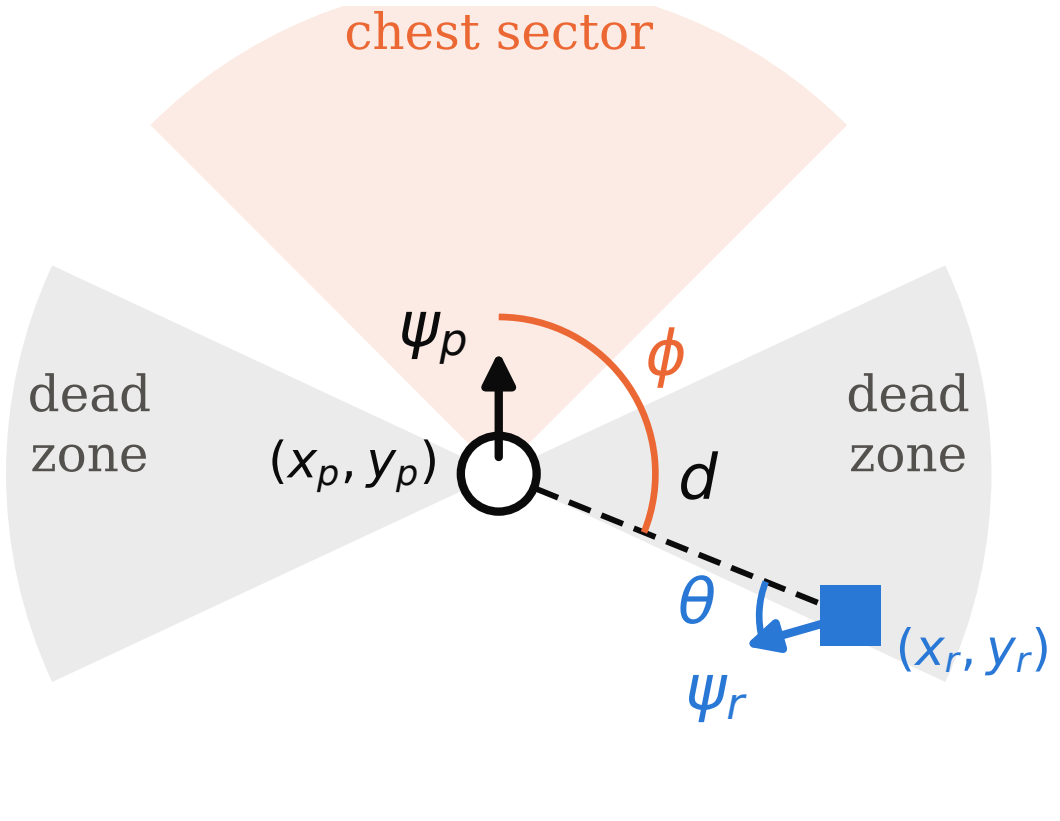
chest sector
dead zone
dead zone
ψ_p
ϕ
(x_p, y_p)
d
θ
(x_r, y_r)
ψ_r